

PV–PORT 180 A port board

Rev F0 | 2026–10–04 | Schematics authored by Chinmoy Bhuyan | Ports A (PV) and B (bus/battery) of PV–P100/110, common negative BUS–

01 Connectors and sensor supply

PORT 26–way to CTRL–C2000, 14–way to SYS–IO–AUX, ID + hold readback, +24V → +3V3S buck, AGND tie, PE bond

File: sheets/01_io.kicad_sch

02 Port A power path

Terminals, fuses, SPD, EMI filter, contactor, precharge relay, coil drives, hold stage, shunt, discharge + time limit, bleeder

File: sheets/02_portA_power.kicad_sch

03 Port A sensing and hold logic

VA bus side, VAX terminal side (AMC3330), IA current (AMC3302), two–channel hold logic, proof–test injection, readback RB1/RB2

File: sheets/03_portA_sense.kicad_sch

04 Port B power path

Terminals, fuses, SPD, EMI filter, contactor, precharge relay, coil drives, hold stage, shunt, discharge + time limit, bleeder

File: sheets/04_portB_power.kicad_sch

05 Port B sensing and hold logic

VB bus side, VBX terminal side (AMC3330), IB current (AMC3302), two–channel hold logic, proof–test injection, readback RB1/RB2

File: sheets/05_portB_sense.kicad_sch

06 Insulation measurement

AUX1 = V(PE–BUS–), AUX2 = V(PE–A+) (AMC3330); 992 k test strings A+ → PE, BUS– → PE, each with a common–source SiC pair (TPSI3050)

File: sheets/06_imd.kicad_sch

Common negative: port A and port B share BUS– (non–isolated buck–boost), so both ports are one HV domain; the main contactors switch the + pole only, fuses are in both poles. Chassis–mounted parts (terminals, fuses, CM choke, contactors, RST 200 resistors, thermostats, shunt) are drawn here with their real MPN or an RFQ / CUSTOM specification.

Isolators (checked on every build): AMC3330/AMC3302, TPSI3050–Q1, CNY65B, contactor and relay coil–contact. Declared HV–PE crossings: MOV3 of each varistor Y, Y capacitors, the PE–side SiC of each insulation–measurement pair, the PE–end element of the AUX dividers.

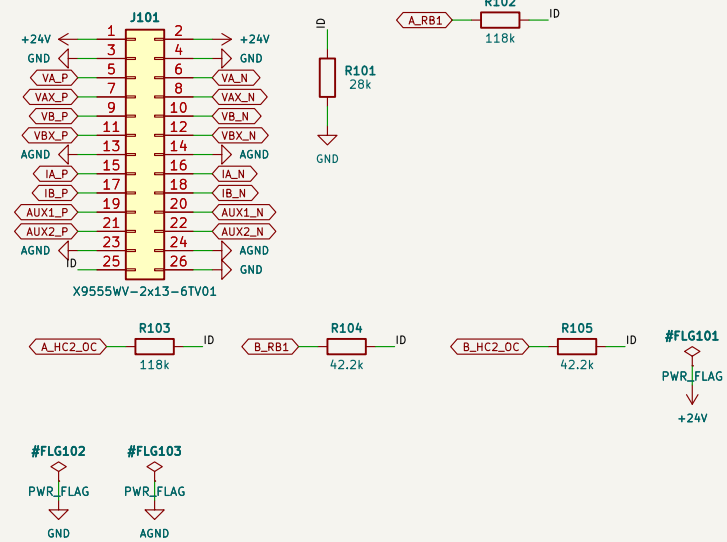
IC–17/IC–18: the whole board is conformal–coated to PD1 (IEC 60664–3 Type 1) over all HV copper and every HV–PELV isolator footprint (8–8.5 mm packages < 10 mm reinforced at PD2). Only the NH fuse links / bases and the DC terminals sit in the cooling–air path (PD3); the board, varistor network and branch fuses are in the closed PD2 compartment.

Surge credit (D–032, IC–01/02/03): the on–board varistor Y limits pole–PE and pole–pole to 3.8 kV at In 3 kA (calculated) and is monitored: reinforced is dimensioned for 6 kV (AMC3302 6.25 kV, AMC3330 7.7 kV pass on this case only). IC–04 OPEN: the G7L coil–contact rating (4000 V AC) is below reinforced – fix in rev F1 (own isolated coil supply).

SYS–IO–AUX: K_DISCH energised first at power–up; the port block itself inhibits the discharge of a closed port. Coil flyback never reaches the DO clamps in normal operation (two–switch coil drives, varistor / zener across each coil).

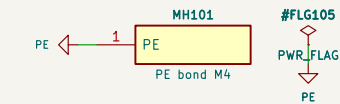
Not bench–validated. Values: sim/port_design.py → port_spec.json		
Sheet 1		
Schematics authored by Chinmoy Bhuyan		
HV–PELV reinforced (1000 VDC, OVC II, PD2, coated to PD1); HV–PE basic + bonding		
Sheet: /		
File: PV–PORT–180.kicad_sch		
Title: PV–PORT 180 A port board – Cover / Hierarchy		
Size: A3	Date: 2026–10–04	Rev: F0
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PORT CONNECTOR (CTRL-C2000) + ID WITH HOLD READBACK



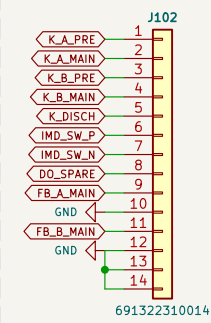
interfaces.PORT, ribbon 1:1. AMC outputs drive the cable directly (≤ 500 pF, AMC3330 p4); anti-alias RC is on CTRL-C2000. ID: 28k || readback network = 9.99 k (PV=PORT code 10.0k; CTRL-C2000 rev D values). RB outputs through 2 x 118k (port A) and 2 x 42.2k (port B): nine states, ≥ 155 counts apart (81 with the hold coils on, sim/port_design.py sec. 7.4); CTRL decodes them with its port-only table.

PE BOND



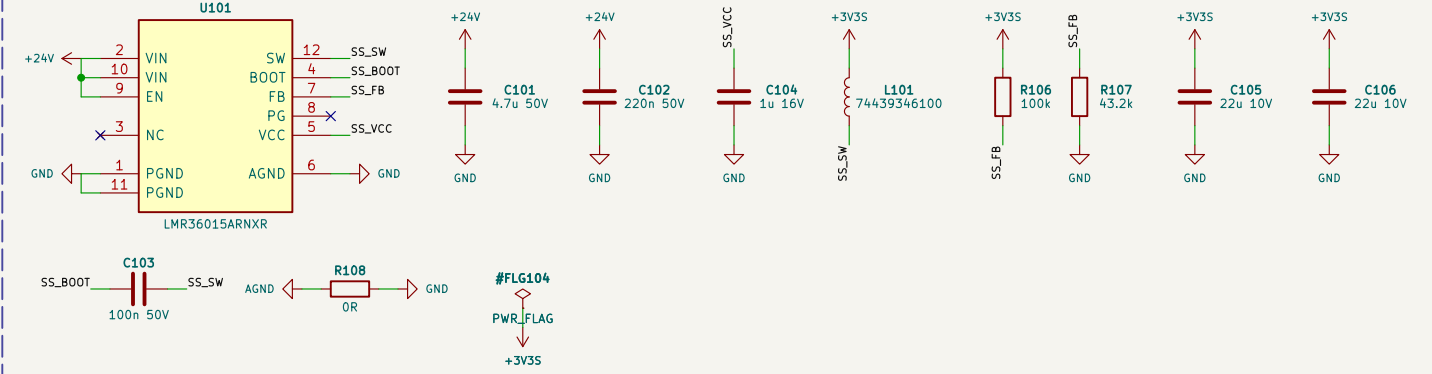
Board PE to chassis at the SPD / Y-capacitor end.

CONTROL CONNECTOR (SYS-IO-AUX)



port.CTRL_PINS: D01-D08 (DO_SPARE = proof test), FB_A + its GND, FB_B + its GND (INT-02), 2 coil returns. Port B (battery) drives 2 HVC43 coils: K_B_MAIN up to 0.60 A -> SYS D04 needs ILIM 10 k (1.58-2.30 A).

SENSOR SUPPLY +3V3S



LMR36015 400 kHz, 24 V -> 3.315 V (100 k / 43.2 k, SNVSB49D sec 10.2), <= 0.34 A for 6 x AMC3330 + 2 x AMC3302 + hold logic. AGND joins GND only here (0 ohm).

Figure 1: Typical connection diagram of the AMC3330DWE. The diagram shows the internal pin connections of the AMC3330DWE (U501) to various external components. The chip has pins for INP, INN, OUTP, OUTN, DIAG, VDD, LDO_OUT, LDO_IN, DCDC_OUT, DCDC_IN, DCDC_HGND, and DCDC_GND. External components include resistors R502-R506, capacitors C502-C509, and a bus connection to the TNPV12061M00BEEA. The diagram also shows connections for +3V3S, +3V3S, and +3V3S to AGND, and a connection for +3V3S to +3V3S.

Figure 10 is a schematic diagram of the TNPV12061M00BEEA board. It shows the internal components of the board, including the AMC3330WE chip, various resistors (R509, R510, R511, R512, R513, R514, C510, C511, C512, C515, C516, C517, C518), capacitors (C511, C512, C515, C516, C517, C518), and connectors (X1, X2, X3, X4, X5, X6, X7, X8, X9, X10, X11, X12, X13, X14, X15, X16, X17, X18, X19, X20, X21, X22, X23, X24, X25, X26, X27, X28, X29, X30, X31, X32, X33, X34, X35, X36, X37, X38, X39, X40, X41, X42, X43, X44, X45, X46, X47, X48, X49, X50, X51, X52, X53, X54, X55, X56, X57, X58, X59, X60, X61, X62, X63, X64, X65, X66, X67, X68, X69, X70, X71, X72, X73, X74, X75, X76, X77, X78, X79, X80, X81, X82, X83, X84, X85, X86, X87, X88, X89, X90, X91, X92, X93, X94, X95, X96, X97, X98, X99, X100). The board is powered by a 1.4V supply and has a 1.4V output. The board is labeled 'U502' and 'AMC3330WE'.

Figure 8-2: Recommended BOM for the AMC3302DWE. The diagram shows the internal pin connections of the AMC3302DWE (U503) to various external components. The chip has pins 1-16. Pins 1-4 are connected to IB_INP, IB_INN, IB_DCO, and IB_HLDO respectively. Pins 5-8 are connected to DCDC_OUT, HLDO_IN, HLDO_OUT, and NC. Pins 9-16 are connected to DCDC_GHND, DCDC_GND, AGND, and various other pins. The diagram also shows the connection of the shunt resistor (R515, R516) and the current sense resistor (R517) to the IB_INP, IB_INN, IB_DCO, and IB_HLDO pins. The shunt resistor is connected to the IB_INP and IB_INN pins, and the current sense resistor is connected to the IB_DCO and IB_HLDO pins. The diagram also shows the connection of the current sense resistor (R517) to the IB_DCO and IB_HLDO pins. The diagram also shows the connection of the current sense resistor (R517) to the IB_DCO and IB_HLDO pins. The diagram also shows the connection of the current sense resistor (R517) to the IB_DCO and IB_HLDO pins.

$|B| > +10 \text{ A}$ (band 382–441 A, own REF3030E + 0.1 % network, CM-cancelling) AND FB = closed AND
 DIAG = valid $\rightarrow$ HOLD1. Test injection 86.6 k from B_TST_T (side A) / B_TST_TD (side B): shifts the threshold
 by 459 A, so a healthy side trips at 0 A under TEST only.

$|I_B| > 410 \text{ A}$ (band 382–444 A, own REF3030E + 0.1 % network CM–cancelling) AND FB = closed AND
 DIAG = valid $\rightarrow$ H0L02: Test Injection 86.6 k Ω from B_TST_T2A (side A) / B_TST_T2B (side B): shifts the threshold
 by 459 A, so a healthy side trips at 0 A under TEST only.

100 k / 15 k $\rightarrow$ Schmitt buffer T (ch1 side A). T $\rightarrow$ 47 k / 22 μ F $\rightarrow$ T_d (ch1 side B). One-shots (3.3 V / 47 k) of T $\rightarrow$ T2A (ch2 side A) and of T_d $\rightarrow$ T2B (ch2 side B). No single fault trips both channels for longer than one one-shot. Firmware runs it at every start, connector opens and closed (report sec. 7.4; windows, what it finds and what it does not).

The schematic diagram illustrates the test setup for current measurement. It features two microcontroller units, U516 and U517, connected to a common ground (AGND). U516 is connected to a +3V3S supply and a 15k resistor (R556) to AGND. U517 is connected to a +3V3S supply and a 100k resistor (R555) to AGND. The output of U516 is connected to the input of U517. The output of U517 is connected to a 100k resistor (R555) to AGND. The output of U517 is also connected to a 100k resistor (R555) to AGND. The output of U517 is also connected to a 100k resistor (R555) to AGND.

Not bench-validated. Values: sim/port_design.py -> port_spec.json

Sheet 6

Schematics authored by Chinmoy Bhuyan

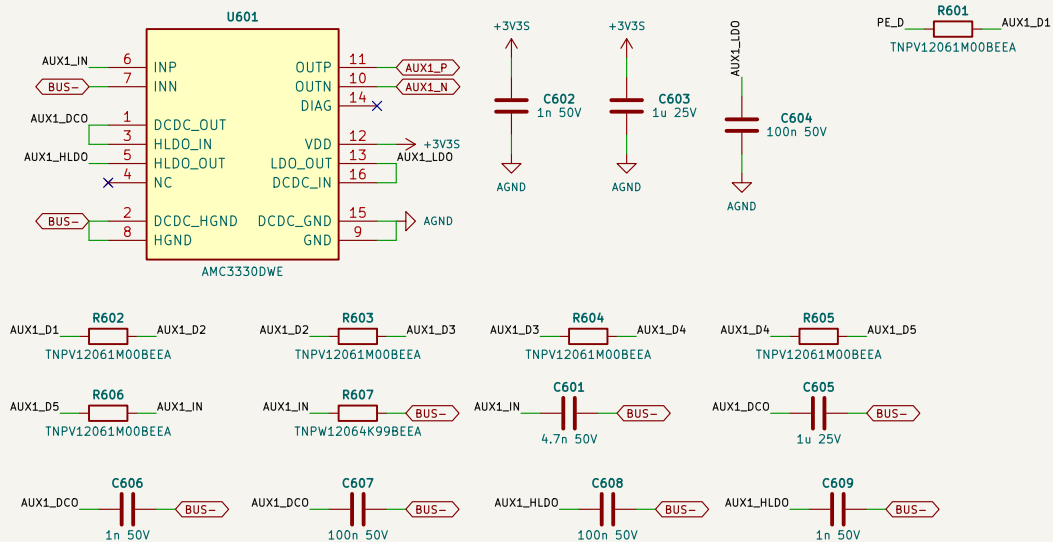
HV-PEM reinforced (1000 VDC, OVC II, PD2, coated to PD1): HV-PE basic + bonding

Sheet: /05 Port B sensing and hold logic/
File: 05_portB_sense.kicad_sch

Title: PV-PORT 180 A port board – Port B sensing and hold logic

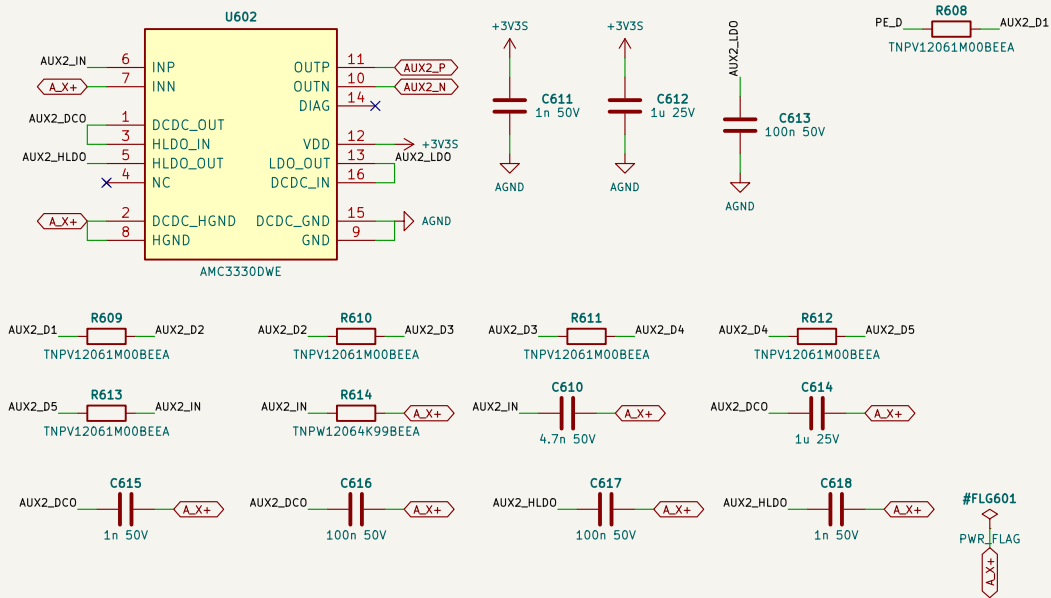
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AUX1 – PE RELATIVE TO BUS–



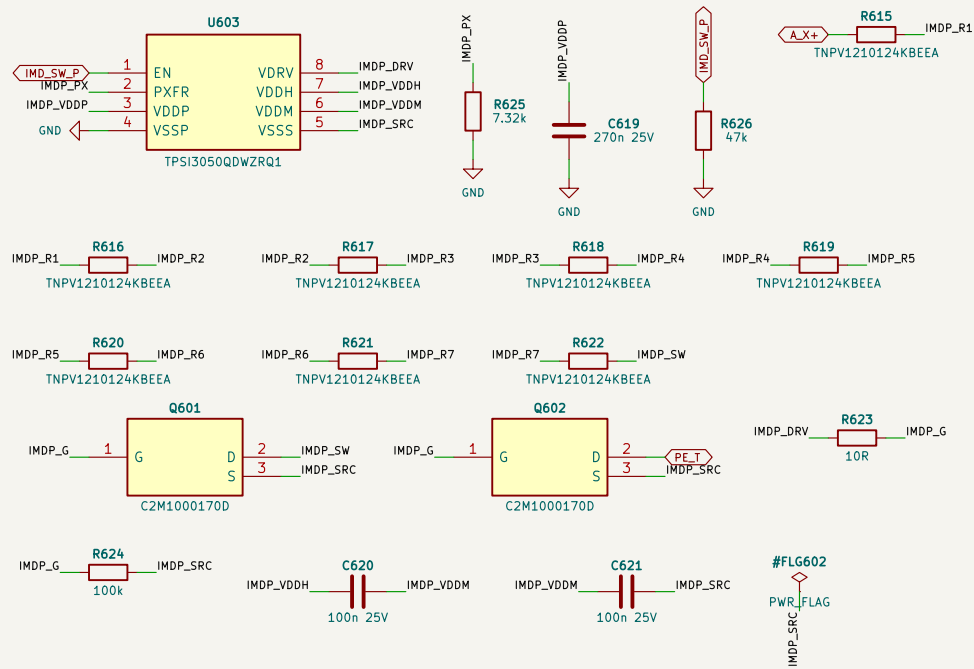
6 x 1.00 M from PE + 4.99 k: +/-1203 V linear. The divider is a fixed, known 6.005 M path to PE (included in the IMD equations, sim/port_design.py sec. 8). First element = HV–PE crossing.

AUX2 – PE RELATIVE TO A_X+



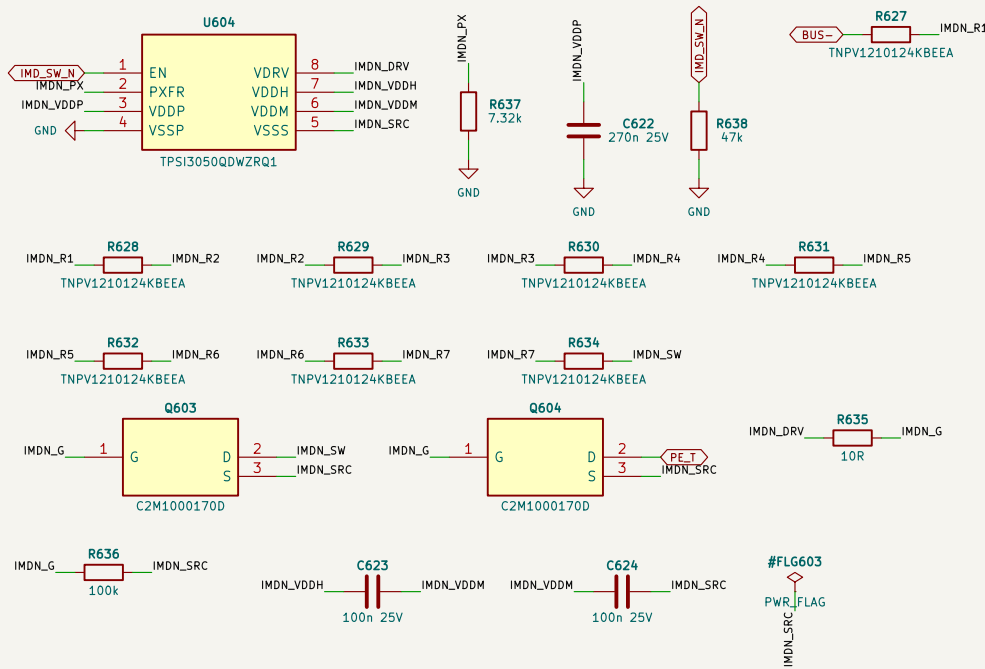
6 x 1.00 M from PE + 4.99 k: +/-1203 V linear. The divider is a fixed, known 6.005 M path to PE (included in the IMD equations, sim/port_design.py sec. 8). First element = HV–PE crossing.

TEST STRING A_X+ -> PE (IMD_SW_P)



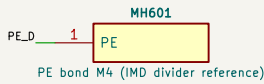
8 x TNPV1210 124 k = 992 k (125 V, 0.13 W each at 1000 V): 1 mA test current. Energised = connected. Common-source C2M1000170D pair (PA–09): blocks either polarity, so a pole below PE cannot leak through a body diode. TPSI3050 two-wire from the 24 V DO, reinforced. IDSS <= 100 uA each – EOL baseline cal.

TEST STRING BUS– -> PE (IMD_SW_N)



8 x TNPV1210 124 k = 992 k (125 V, 0.13 W each at 1000 V): 1 mA test current. Energised = connected. Common-source C2M1000170D pair (PA–09): blocks either polarity, so a pole below PE cannot leak through a body diode. TPSI3050 two-wire from the 24 V DO, reinforced. IDSS <= 100 uA each – EOL baseline cal.

PE_D BOND + ONE ACTIVE MONITOR PER DC SYSTEM



PE_D: own M4 bond (dividers only). The module disables its own measurement by leaving IMD_SW_P/N de-energised (strings open, fail-safe): a passive module still loads each pole with 6.005 M to PE.